

2. What is Marble?

- (a) Sedimentary rock
- (b) Felsite rock
- (c) Metamorphic rock
- (d) Igneous rock

U.P.P.C.S. (Pre) 1991

Ans. (c)

Marble is a metamorphic rock that is formed when limestone is subjected to heat and pressure of metamorphism.

3. Marble is –

- (a) Re-crystallized limestone
- (b) An igneous rock
- (c) Sandstone
- (d) Formed from changes of organic matter to inorganic ones

Uttarakhand P.C.S. (Pre) 2006

Ans. (a)

Marble is a metamorphic rock. It is composed of re-crystallized carbonate minerals.

D. Energy Mineral

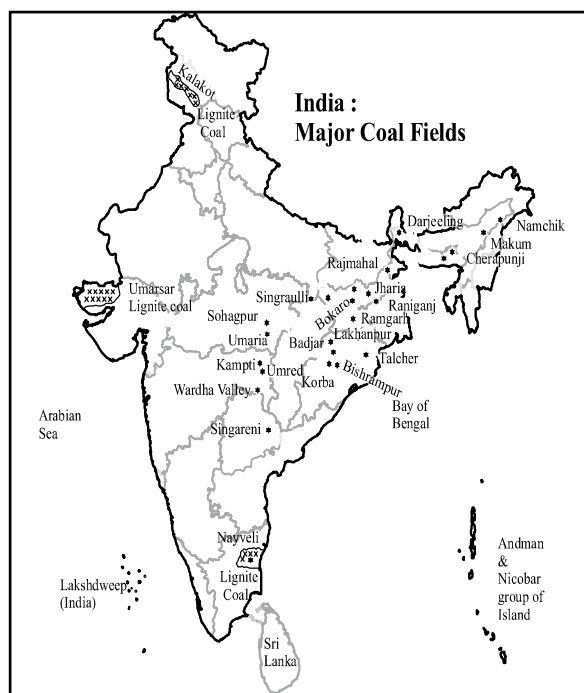
i. Coal

*Coal is solid sedimentary rock mainly constituted of hydrocarbons. Coal is mainly used as a fuel producing heat and energy. It is remains of organic matter and animal fossils accumulated in sedimentary rock by a different methods.

*Coal occurs in rock series of two main geological ages-

(1) Gondwana (2) Tertiary

*Gondwana coal is **high quality coal** with **low ash content**. while tertiary coal is **low quality coal**. Most of the coal produced in India is obtained from this region. Bituminous coal is abundantly found here. It has low sulfur content. According to data from Indian Minerals Yearbook 2022, in the coalfields of India, most coalfields are related to Lower Gondwana. More than about 99% of coal (as of April, 2022) deposits and production have been covered by Gondwana coal. *Major Coalfields of Gondwana System are - **Damodar Valley** Coalfield, **Son Valley** Coalfield, Mahanadi Valley Coalfield, **Godavari** Coalfield, **Satpura** Coalfield, **Vardha** Ghati Coalfield and **Rajmahal** Coalfield.



***Raniganj** coalfield of West Bengal is situated in the upper Damodar valley which is the largest and the most important coalfield of India.

*Coal is found in a variety of forms depending on the degree of compression and the depth and time of burial. *There are four major categories of coal based on the carbon content and impurity –

- (1) Peat
- (2) Lignite
- (3) Bituminous
- (4) Anthracite

*Carbon content in **peat** is **less than 25%**. *Carbon content in lignite coal is up to 25-35%. Carbon content in bituminous coal is up to 45-86%. Carbon content in anthracite coal is up to 86-97%. ***Neyveli** in Tamil Nadu is a principal lignite reserve. *Coal that has been buried deep and subjected to increased temperature is **bituminous** coal. It is the most popular coal in commercial use. ***Anthracite** is the high quality hard coal. *According to data from Indian Mineral Yearbook 2022, **coal reserves/resources** (as of April 1, 2022) are found chronologically in Odisha (88105 million tonnes), Jharkhand (86660 million tonnes), Chhattisgarh (74192 million tonnes), West Bengal (33871 million tonnes), and Madhya Pradesh (30917 million tonnes).

*The total production of coal reported in 2021-22 (P) was 778.21 million tonnes, which increased by around 9% in

comparison to that of the previous year. Odisha is the largest coal producing state with a share of about 24%, followed by Chhattisgarh and Madhya Pradesh, having a contribution of 20% and 18% respectively in the national output. Next in order of share in the production were Jharkhand (16.71%), Telangana (8.64%), Maharashtra (7.26%), West Bengal (3.74%) and Uttar Pradesh (2.32%).

*The share of coking coal in the total coal (2021-22) produced in India is 6% and the rest 94% is non-coking coal.

*Major coal fields of Jharkhand are Jharia, North Karnapura, South Karnapura, Bokaro, Chandrapura, Rajmahal, Giridih, etc.

***Jharia coalfield** is situated in Dhanbad district of Jharkhand. Major coking coal deposits are situated here. It is mainly spread across the Chota Nagpur plateau, Jharkhand, Chhattisgarh, Odisha and West Bengal. *Total coal production of Chhattisgarh in 2021-22 (P) was 154120 thousand tonnes and total storage is 74192 million tonnes.

***Korba coalfield** is situated in the Hasdev river basin in Chhattisgarh.

Korba coalfield is an important coalfield of Chhattisgarh.

*Coal reserves of Chhattisgarh are found in **Tatapani-Ramkola, Bistrampur, Sonhat, Jhilimili, Chirimiri, Lakhanpur, Panchbahini and Sendurgarh** etc.

***Talcher** is a major coalfield of Odisha situated in the valley of the Brahmani River in Angul district. Talcher coalfield is managed by **Mahanadi coalfield Limited** which is a subsidiary of Coal India Limited.

*Raniganj is a coalfield of West Bengal. Coalfields of West Bengal are **Barjora, Birbhum and Darjeeling**.

***Namchik- Namphuk and Miao Bum coalfields** are situated in Arunachal Pradesh.

***Coalbed methane** and **shale gas** are unconventional sources of energy. Coalbed methane is 90% more pure methane gas extracted from the coal seam. Shale gas is extracted from fine-grained sedimentary rocks.

*Shale gas is not only a mixture of propane and butane but it also contains more than 90% methane, 4% ethene, 1% propane and various other gases. Huge **reserves of shale gas** are found in the **Sindhu-Gangetic region, Assam, Gujarat, Rajasthan** and the **coastal region**. Currently, coal industry in India is facing a lot of challenges. Some of them are **low-**

quality coal, deficiencies in coal washing establishment, higher imports of coking coal, etc.

*Coal (proved) reserves found in major states (April 1, 2022) in chronological order are-

- (1) Jharkhand (53245 Million Tonnes)
- (2) Odisha (48573 Million Tonnes)
- (3) Chhattisgarh (32053 Million Tonnes)
- (4) West Bengal (17234 Million Tonnes)
- (5) Madhya Pradesh (14052 Million Tonnes)

1. Where was first coal mine in India mined?

- (a) Jharia
- (b) Raniganj
- (c) Dhanbad
- (d) Asansol
- (e) None of the above / More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (b)

In India, the first coal mine was mined on the banks of the Damodar River at Raniganj. It was first mined in 1774 AD.

2. Consider the important events of coal mining in India and arrange these in correct chronological order starting from the earliest to the last activity :

I. First production of coal in Raniganj

II. Establishment of Coal India Ltd. (CIL)

III. Nationalisation of coal mines

IV. Establishment of National Coal Development Corporation (NCDC)

Select the correct answer using the code given below :

Code :

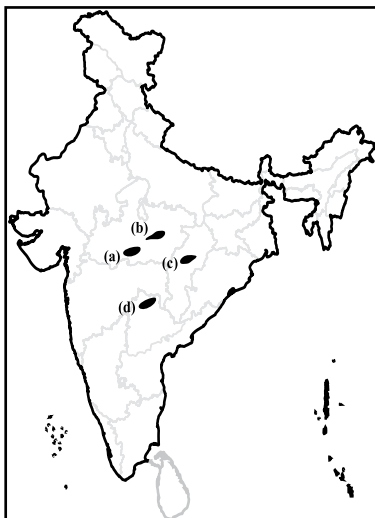
- (a) I, II, III, IV
- (b) III, I, II, IV
- (c) I, IV, III, II
- (d) I, III, II, IV

U.P. R.O./A.R.O. (Pre) 2023

Ans. (c)

India has a long history of commercial coal mining covering nearly 220 years starting from 1774 by M/s Sumner and Heatly of the East India Company in the Raniganj coal field along the western bank of river Damodar. The National Coal Development Corporation (NCDC) was established by the Government of India in the year 1956. The nationalization was done in two phases : the first with the coking Coal Mines in 1971-72 and then with the non-coking coal mines in 1973. Another enactment, namely the coal mines (Taking Over the Management) Act, 1973, extended the right of the Government of India to take over the management of the coking and non coking coal mines in seven states including the coking coal mines taken over in 1971.

3. On the outline map of India (a), (b), (c) and (d) depict the location of four coal fields. Which one of the following is not correctly shown on the map?



- (a) Satpura (b) Son Valley
(c) Korba (d) Karanpura

Uttarakhand P.C.S. (Pre) 2012

Ans. (d)

According to the above map, option (d) is depicted wrong as Karanpura is a coalfield located in the state of Jharkhand. The location of all other places in the map is approximately correct.

4. Which one of the following rock systems in India is the main source of coal deposits?

- (a) Dharwar system (b) Gondwana system
(c) Cuddapah system (d) Vindhyan system

U.P.P.C.S. (Pre) 2008

U.P. U.D.A./L.D.A. (Pre) 2006

Ans. (b)

The coal deposits in India are primarily concentrated in the Gondwana sediments occurring mainly in the eastern and central parts of Peninsular India, although Gondwana coal deposits also occur in Assam and Sikkim in northeastern part of the country. According to data from Indian Minerals Yearbook 2022, more than about 99% of coal (as of April 1, 2022) deposits and production have been covered by Gondwana coal.

5. Which of the following rock systems provides over 90% of the coal of India?

- (a) Vindhyan system (b) Dharwar system
(c) Tertiary system (d) Gondwana system

U.P.P.C.S. (Pre) 2017

Ans. (d)

See the explanation of the above question.

6. Which of the following systems of rocks provides maximum amount of coal in India?

- (a) Dharwad System (b) Gondwana System
(c) Vindhyan System (d) Tertiary System

U.P. R.O./A.R.O. (Pre) (Re-Exam) 2016

Ans. (b)

See the explanation of the above question.

7. Which of the following is/are the characteristic/characteristics of Indian coal?

1. High ash content
2. Low sulphur content
3. Low ash fusion temperature

Select the correct answer using the code given below –

- (a) 1 and 2 (b) Only 2
(c) 1 and 3 (d) All of these

I.A.S. (Pre) 2013

Ans. (a)

Coal in India is found exclusively in two areas : (i) Gondwana region and (ii) Kalp region. The coal of Gondwana region is of high grade. The quantity of ash is less and thermogenic power is high whereas the coal of the third Kalp region is of inferior quality. It is rich in sulfur and therefore not used for certain industries. According to data of Indian Minerals Yearbook 2022 (as on April 1, 2022), most of India's coal is obtained (more than about 99%) from the Gondwana region. Most bituminous coal is found here; it has a low amount of sulfur.

8. Which one of the following is the appropriate reason for considering the Gondwana rocks as most important of rock systems of India?

- (a) More than 90% of limestone reserves of India are found in them
(b) More than 90% of India's coal reserves are found in them
(c) More than 90% of fertile black cotton soils are spread over them
(d) None of the reasons given above is appropriate in this context

I.A.S. (Pre) 2010

Ans. (b)

According to the data of Indian Minerals Yearbook 2022 (as on April 1, 2022), more than about 99% of coal reserves of India are found in the Gondwana region.

9. At the present state of our knowledge and resource position, India will remain self-sufficient for the next three decades in –

- (a) Tin (b) Coking coal
(c) Copper (d) Petroleum

I.A.S. (Pre) 1993

Ans. (b)

According to the Energy Policy report, (Plannig Commission), the petroleum reserves in India will be depleted in 22 years, gas reserves in 30 years and coal reserves in 80 years. According to Indian Minerals Yearbook 2022, about 361411 million tonnes of coal resources up to 1200m depth have been established in the country as of April 1, 2022. Of the total resource share, the share of prime coking coal is 5318 million tonnes, medium-coking is 28080 million tonnes & semi-coking is 1708 million tonnes and non-coking coal is 324651 million tonnes.

10. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R)?

Assertion (A) : According to Geological Survey of India the total reserve of all grade coal in India is 293.50 billion tonnes.

Reason (R) : More than half of total proved reserve of coal of the country is found in two state - Jharkhand and Odisha.

Choose the correct answer from the codes given below:

Codes :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is the correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true.

U.P. B.E.O. (Pre) 2019

Ans. (*)

According to data of Indian Minerals Yearbook 2022 (as on 1 April, 2022), of the coal reserves in India are 361411 million tonnes. And More than half of the total proved reserves of coal in the country are found in two states - Jharkhand and Odisha. The Public Service Commission has considered the answer to this question as an option (b), while (R) is correct. Hence the correct answer will be an option (d).

11. Which of the following two states of India, have more than 50 percent of the total country's coal reserves ?

- (i) Jharkhand (ii) Madhya Pradesh
(iii) Odisha (iv) Chhattisgarh
(a) (i) and (ii) (b) (i) and (iii)
(c) (iii) and (iv) (d) (ii) and (iii)

M.P.P.C.S. (Pre) 2024

Ans. (b)

According to the data of Indian Minerals Yearbook 2022, as of 1 April 2022, the states with the highest geological coal resources in the country are Odisha (88105 million tonnes), Jharkhand (86,660 million tonnes) and Chhattisgarh (74,192 million tonnes). During the same period, the total geological coal resources in India were estimated at 361411 million tonnes. Thus, Odisha and Jharkhand together account for 48.4% of the country's total coal deposits. Therefore, option (b) can be considered the correct answer.

12. Consider the following statements with reference to coal :

1. India is one of the world's leading producers of coal.
2. India has vast coal reserves in states like Jharkhand and Chhattisgarh.

Select the correct answer from the code given below:

Code :

- (a) Only 2 (b) Both 1 and 2
(c) Neither 1 nor 2 (d) Only 1

U.P. R.O./A.R.O. (Pre) 2023

Ans. (b)

According to the Indian Minerals Yearbook, 2022, China is the largest producer of coal and lignite with about 50% share in the total world production of coal in the year 2021. It is followed by India (9.0%), United States of America (7.88%), Indonesia (7.80%), Australia (6.02%) and Russia (5.48%) respectively. As on April 1, 2022 Odisha, Jharkhand and Chhattisgarh are major states that have the largest coal reserves/resources.

13. The descending order of the states in coal production is-

- (a) Bihar, Madhya Pradesh, West Bengal
(b) Madhya Pradesh, West Bengal, Bihar
(c) West Bengal, Madhya Pradesh, Bihar
(d) Bihar, West Bengal, Madhya Pradesh

43rd B.P.S.C. (Pre) 1999

Ans. (a)

In the year 1999 (when the question was asked), the production of coal in undivided Bihar (including Jharkhand) was more than that of undivided Madhya Pradesh (including Chhattisgarh). Thus, the correct answer was option (a). At present [2021-22(P)], the three largest coal-producing states in India are : 1. Odisha, 2. Chhattisgarh, 3. Madhya Pradesh.

14. Two leading coal producing states in India are –

- (a) Jharkhand and Odisha
- (b) Jharkhand and Chhattisgarh
- (c) Chhattisgarh and Madhya Pradesh
- (d) Madhya Pradesh and Andhra Pradesh

U.P. U.D.A./L.D.A. (Pre) 2010

Ans. (b)

Option (b) is the correct answer when the question was asked. According to the data of the Indian Minerals Yearbook 2022, two leading coal producing (2021-22, P) states are Odisha & Chhattisgarh.

15. Arrange the following states in descending order of coal production and choose the correct answer using code given below:

- | | |
|-----------------------------|-----------------------|
| (i) Chhattisgarh | (ii) Jharkhand |
| (iii) Madhya Pradesh | (iv) Odisha |

Code :

- | | | | |
|-----------|------|-------|-------|
| (a) (iv) | (i) | (ii) | (iii) |
| (b) (ii) | (iv) | (iii) | (i) |
| (c) (ii) | (i) | (iv) | (iii) |
| (d) (iii) | (ii) | (iv) | (i) |

U.P.P.C.S. (Pre) 2008

Ans. (c)

Coal, designated as "Black Diamond", is one of the main sources of energy. It is also known as "Fuel Mineral". Chemically, it contains carbon which gives it not only the color but also determines its thermal capacity. According to the data for the year 2007-08, the top coal producing states were - Jharkhand, Chhattisgarh, Odisha and Madhya Pradesh. According to the 'Indian Minerals Yearbook 2022, the top 5 coal producing states in 2021-22 (P) were – Odisha> Chhattisgarh> Madhya Pradesh> Jharkhand> Telangana.

16. Which one of the following states is the largest producer of coal in India?

- (a) Andhra Pradesh
- (b) Madhya Pradesh

- (c) Odisha
- (d) West Bengal

U.P.P.C.S. (GIC) 2010

U.P.P.C.S. (Mains) 2004

U.P.P.C.S. (Mains) 2003

Ans. (c)

According to the Indian Minerals Yearbook 2022, Odisha is the largest coal producing state with a share of about 24%, followed by Chhattisgarh.

17. Which state is the largest producer of coal in India?

- | | |
|--------------------|--------------------|
| (a) West Bengal | (b) Odisha |
| (c) Andhra Pradesh | (d) Madhya Pradesh |

U.P.R.O./A.R.O. (Pre) 2014

Ans. (b)

See the explanation of the above question.

18. Which of the following is the correct sequence of States of India in descending order in terms of reserves of coal?

- (a) Odisha, Chhattisgarh, Jharkhand, Madhya Pradesh
- (b) Chhattisgarh, Odisha, Madhya Pradesh, Jharkhand
- (c) Jharkhand, Odisha, Chhattisgarh, Madhya Pradesh
- (d) Madhya Pradesh, Chhattisgarh, Jharkhand, Odisha

U.P.P.C.S. (Mains) 2014

Ans. (*)

The descending order in terms of coal (Proved) reserves according to data of the Indian Minerals Yearbook 2022 (as on April 1, 2022), – Odisha> Jharkhand > Chhattisgarh > West Bengal > Madhya Pradesh.

19. In which of the following states of India has major reserves of coal ?

- | | |
|---------------|-------------------|
| (a) Jharkhand | (b) Chhattisgarh |
| (c) Odisha | (d) All the above |

70th B.P.S.C. Pre-2024

Ans. (d)

According to the Indian Minerals Yearbook, 2022, the states with the highest total coal reserves in India (as of April 1, 2022) are:

1. Odisha (88,105 million tonnes)
2. Jharkhand (86,660 million tonnes)
3. Chhattisgarh (74,192 million tonnes).

Therefore, option (d) is the correct answer.

20. Which of the following is the largest coal producing state of India?

- | | |
|------------------|---------------|
| (a) Chhattisgarh | (b) Jharkhand |
|------------------|---------------|

- (c) Madhya Pradesh (d) Odisha

U.P.P.C.S. (Pre) 2009

Uttarakhand U.D.A./L.D.A. (Pre) 2006

U.P. Lower Sub. (Pre) 2002

Ans. (d)

According to data of Indian Minerals Yearbook 2022, Odisha (2021-22, P) is the largest Coal producing state in India.

21. Arrange the following states of India in descending order of their coal reserves and select the correct answer from the code given below:

1. Chhattisgarh 2. Jharkhand
3. Madhya Pradesh 4. Odisha

Code :

- (a) 2, 4, 1, 3 (b) 2, 1, 4, 3
(c) 1, 4, 3, 2 (d) 3, 2, 1, 4

U.P.P.C.S. (Mains) 2010

Ans. (a)

When the question was asked option (a) was the correct answer. Coal (proved) reserves found in major states (1 April, 2021) in chronological order are -
Jharkhand > Odisha > Chhattisgarh > West Bengal > Madhya Pradesh.

22. The correct sequence in descending order of the following states in terms of the proved coal reserves, as per the report of the Ministry of Coal as on 1-4-2014 is:

- (a) Chhattisgarh, Odisha, Jharkhand, Madhya Pradesh
(b) Odisha, Chhattisgarh, Madhya Pradesh, Jharkhand
(c) Jharkhand, Chhattisgarh, Odisha, Madhya Pradesh
(d) Jharkhand, Odisha, Chhattisgarh, Madhya Pradesh

U.P.R.O./A.R.O. (Pre) 2016

Ans. (d)

The correct sequence in descending order as per the report of the Indian Minerals Yearbook 2022, Coal (Proved) on April 1, 2022, is : Jharkhand (53245 million tonnes), Odisha (48573 million tonnes), Chhattisgarh (32053 million tonnes), West Bengal (17234 million tonnes) and Madhya Pradesh (14052 million tonnes).

23. The correct sequence of the following coal producing States in terms of production (2014) in descending order is –

- (a) Chhattisgarh, Jharkhand, Odisha, Maharashtra
(b) Odisha, Jharkhand, Chhattisgarh, Maharashtra

- (c) Jharkhand, Chhattisgarh, Maharashtra, Odisha

- (d) Jharkhand, Odisha, Chhattisgarh, Maharashtra

U.P.P.C.S. (Mains) 2016

Ans. (a)

According to Coal Statistics 2014-15, Chhattisgarh registered the highest coal production of (22.5% share), followed by Jharkhand (20.0% share) and Odisha (19.9% share). According to the Indian Minerals Yearbook 2022, the top 5 coal producing states in 2021-22 (P) are- Odisha, > Chhattisgarh, > Madhya Pradesh, > Jharkhand, > and Telangana.

24. Growth of Chota Nagpur industrial region is linked with the discovery of :

- (a) Bauxite (b) Coal
(c) Iron ore (d) Manganese

U.P.P.C.S. (Spl.) (Mains) 2004

Ans. (b)

Chota Nagpur industrial region is extended in Jharkhand, North Odisha and West Bengal and is famous for its heavy metallurgical industry. The discovery of coal in the Damodar Valley is the main reason for the development of this region.

25. Jharkhand's share in total coal production in the country, is :

- (a) 40% (b) 45%
(c) 47% (d) 49%

Jharkhand P.C.S. (Pre) 2011

Ans. (*)

According to the data Indian Minerals Yearbook 2022, Jharkhand's share in total coal production in the year 2021-22 (P) was 16.72%. None of the options are correct.

26. Three leading producers of coal in descending order are–

- (a) Chhattisgarh, Odisha and Jharkhand
(b) Jharkhand, Chhattisgarh and Odisha
(c) Odisha, Chhattisgarh and Jharkhand
(d) Chhattisgarh, Madhya Pradesh and Andhra Pradesh

U.P.P.C.S. (Mains) 2009

Ans. (a)

According to the data for the year 2008-09, the maximum share in coal production was of Chhattisgarh (20.7%) followed by Odisha (20.0%) and Jharkhand (19.5%). As per Indian Minerals Yearbook 2022, the share of the state in coal production (2021-22, P) was Odisha (24%), Chhattisgarh (20%) and Madhya Pradesh (18%).

27. In India two largest producers of coal (2008-09) are –

- (a) Andhra Pradesh and Chhattisgarh
- (b) Chhattisgarh and Odisha
- (c) Chhattisgarh and Jharkhand
- (d) Jharkhand and Madhya Pradesh

U.P.P.C.S. (Pre) 2011

Ans. (b)

See the explanation of the above question.

28. The Korba coalfield is located in –

- (a) Odisha
- (b) Chhattisgarh
- (c) West Bengal
- (d) Assam

44th B.P.S.C. (Pre) 2000

Ans. (b)

The Korba coalfield is located in Korba district of Chhattisgarh in the basin of the Hasdeo River.

29. Match List-I with List-II and select the correct answer from the codes given below the list :

List- I (Coal producing Areas)	List- II (Coalfields)
A. Damodar Valley	1. Talcher
B. Son Valley	2. Barakar
C. Godavari Valley	3. Umaria
D. Mahanadi Valley	4. Singareni

Code :

A	B	C	D
(a) 2	3	4	1
(b) 3	2	1	4
(c) 1	3	4	2
(d) 4	1	2	3

I.A.S. (Pre) 2010

Ans. (a)

The correct match is as follows :

(Coal producing Areas)	(Coalfields)
Damodar Valley	Barakar
Son Valley	Umaria
Godavari Valley	Singareni
Mahanadi Valley	Talcher

30. Match List- I with List- II and select the correct answer using the code given below the lists :

List- I (Coal Fields)	List- II (Located)
A. Talcher	1. Damodar Valley
B. Karnapura	2. Son Valley
C. Singroli	3. Godavari Valley
D. Singreni	4. Mahanadi Valley

Code :

	A	B	C	D
(a)	4	1	2	3
(b)	3	2	1	4
(c)	2	4	3	1
(d)	1	3	2	4

U.P.P.C.S. (Pre) 2019

Ans. (a)

List- I (Coal Fields)	List- II (Located)
Talcher	- Mahanadi Valley
Karnapura	- Damodar Valley
Singroli	- Son Valley
Singreni	- Godavari Valley

So, as per option (a) is the correct answer.

31. Talcher is a famous coal region, located in :

- (a) Andhra Pradesh
- (b) Chhattisgarh
- (c) Jharkhand
- (d) Odisha

U.P.P.S.C. (GIC) 2010

Ans. (d)

Talcher is a prominent coal region in Angul district of Odisha. The management of the coal region in Talcher is in the hands of Mahanadi Coalfield Ltd., a subsidiary of Coal India Limited.

32. Talcher is the famous coal field of :

- (a) Madhya Pradesh
- (b) Chhattisgarh
- (c) Bihar
- (d) Odisha

M.P.P.C.S. (Pre) 2015

Ans. (d)

See the explanation of the above question.

33. The contribution of Chota Nagpur in production of coal, is approx :

- (a) 20%
- (b) 40%
- (c) 60%
- (d) 80%

U.P.P.C.S. (Pre) 1998

Ans. (d)

When the question was asked option (d) was the correct answer. Chota Nagpur plateau is extended mainly in Jharkhand, Chhattisgarh, Odisha and West Bengal. It has a 64.04% [2021-22 (P)] contribution to India's total coal production.

34. Coal mines in Jharkhand are located at :

- (a) Jharia
- (b) Jamshedpur

(c) Ranchi

(d) Lohardaga

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (a)

Jharia coalfield is located in Dhanbad district of Jharkhand. Most of the coking coal reserves are found here. Jamshedpur is famous for the iron/steel industry; Lohardaga is famous for the production of bauxite and Ranchi is famous for heavy machines.

35. In which of the following coal region the coal reserve is maximum?

(a) Jharia

(b) Raniganj

(c) Korba

(d) Singrauli

U.P.P.C.S. (Pre) 1999

Ans. (b)

According to the Indian Minerals Yearbook 2015, total coal reserves in the aforesaid region are as follows –

Field	Proved Reserves (Million Tonnes)	Total Reserves (Million Tonnes)
Jharia (Jharkhand)	15127.97	19430.06
Raniganj (Jharkhand)	1538.19	2036.30
Raniganj (West Bengal)	13317.36	24632.83
Korba (Chhattisgarh)	5651.14	11755.66
Singrauli (Madhya Pradesh)	6532.68	14803.52
Singrauli (Uttar Pradesh)	884.04	1061.86

According to the Indian Minerals Yearbook 2022, total coal reserves are given as follows (April 1, 2022) -

State	Proved Reserves (Million Tonnes)	Total Reserves (Million Tonnes)
Jharia (Jharkhand)	16653	19533
Raniganj (Jharkhand)	1594	2039
Raniganj (West Bengal)	16815	26357
Korba (Chhattisgarh)	7055	12976
Singrauli (Madhya Pradesh)	8861	16919
Singrauli (Uttar Pradesh)	884	1062

36. What is chiefly found at Jharia in Jharkhand?

(a) Thorium

(b) Silk

(c) Gold

(d) Coal

56th to 59th B.P.S.C. (Pre) 2015

Ans. (d)

See the explanation of the above question.

37. Jharia coalfield is located in which of the following states?

(a) West Bengal

(b) Odisha

(c) Jharkhand

(d) Chhattisgarh

U.P.R.O./A.R.O. (Mains) 2014

Ans. (c)

See the explanation of the above question.

38. Assertion (A): Lignite is an inferior variety of coal containing 35-40 percent of carbon.

Reason (R) : Jharkhand is the largest producer of lignite in India.

In the context of the above statements, which of the following is true:

Code :

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(c) (A) is true, but (R) is false.

(d) (A) is false, but (R) is true.

U.P.U.D.A./L.D.A. (Pre) 2002

Ans. (c)

Lignite is an inferior variety of coal containing 40-55 percent of carbon. Lignite coal is mainly produced (2021-22, P) in three states - Tamil Nadu, Gujarat and Rajasthan. Neyveli is the lignite field in Tamil Nadu and it is the lignite coal mine of India.

39. Largest deposit of lignite coal is found in India at –

(a) Jharkhand

(b) Odisha

(c) Jammu and Kashmir

(d) Tamil Nadu

U.P.U.D.A./L.D.A. (Pre) 2013

Ans. (d)

During the year 2021-22 (P), the reported production of lignite was 47 million tonnes which increased by around 25% in comparison to that of the previous year. The production from Tamil Nadu accounted for around 50%. The production in lignite Gujarat was around 28% and that of Rajasthan was 22%.

40. Which one of the the following is the largest lignite coal field in India?

(a) Singrauli

(b) Makum

(c) Karanpura

(d) Neyveli

U.P. B.E.O. (Pre) 2019

Ans. (d)

Neyveli, located in Tamil Nadu, has the largest reserves of lignite coal in India. According to data of Indian Minerals Yearbook 2022, Tamil Nadu tops in both reserves/resources and production of lignite coal.

41. Which State has the largest reserves of Lignite Coal

- (a) Gujarat (b) Kerala
(c) Rajasthan (d) Tamil Nadu

U.P.P.C.S. (Mains) 2011

Ans. (d)

According to the Minerals Yearbook 2022, there are a total of 46204.16 million tonnes of reserves/resources of Lignite coal in India.

42. Assertion (A) : The inter-state movement of coal forms a major component of inter-state traffic handled by railways.

Reason (R) : Bengal-Jharkhand coalfields are a major source of coal supply to the northwestern states.

Select the correct answer from the code given below:

Code :

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P.U.D.A./L.D.A. (Pre) 2010

Ans. (a)

Bengal-Jharkhand coalfields are a major source of coal supply to northwestern states and the supply of inter-state coal is handled by the railway. They form a major component of handling inter-state traffic. Thus, both (A) and (R) are true, and (R) is the correct explanation of (A).

43. Birsampur is famous for the mining of –

- (a) Copper ore (b) Iron ore
(c) Coal (d) Manganese

U.P.P.C.S. (Pre) 2008

U.P. Lower Sub. (Pre) 2004

Ans. (c)

Birsampur in Surajpur district of Chhattisgarh is famous for coal mining.

44. Match List-I with List-II and choose the correct answer using the codes given below :

List- I (Coalfields)	List- II (States)
A. Karanpura	1. Chhattisgarh
B. Singareni	2. Tamil Nadu
C. Neyveli	3. Jharkhand
D. Korba	4. Andhra Pradesh

Code:

	A	B	C	D
(a)	4	1	3	2
(b)	1	2	4	3
(c)	3	4	2	1
(d)	2	3	1	4

U.P.U.D.A./L.D.A. (Spl.) (Mains) 2010

Ans. (c)

The correct match of coalfields with their respective States is as follows :

Karanpura	-	Jharkhand
Singareni	-	Andhra Pradesh (now in Telangana)
Neyveli	-	Tamil Nadu
Korba	-	Chhattisgarh

45. Despite having large reserves of coal, why does India import millions of tonnes of coal?

1. It is the policy of India to save its own coal reserves for the future, and import it from other countries for the present use.
2. Most of the power plants in India are coal-based and they are not able to get sufficient supplies of coal from within the country.
3. Steel companies need a large quantity of coking coal which has to be imported.

Which of the statement(s) given above is/are correct?

- (a) Only 1 (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3

I.A.S. (Pre) 2012

Ans. (b)

Indian steel companies require a huge amount of coking coal. In order to fulfill that requirement, a large amount of good quality of coal is imported which is not available in India. Most of the electric power plants in India are coal-based. Hence, the internal supply of coal is not properly fulfilled. This is one more reason that coal is imported.

46. With reference to two non-conventional energy sources called 'coalbed methane' and 'shale gas', consider the following statements:

1. Coalbed methane is the pure methane gas extracted from the coal seam, while shale gas is a mixture of propane and butane only that can be extracted from fine-grained sedimentary rocks.
2. In India, abundant coalbed methane sources exist, but so far no shale gas sources have been found.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2014

Ans. (d)

Coalbed Methane and Shale Gas are unconventional sources of energy. Coalbed Methane is 90% more pure methane gas extracted from the coal seam. Shale gas is extracted from fine-grained sedimentary rocks. Shale gas is not only a mixture of Propane and Butane but it also contains more than 86% of Methane, 4% Ethane, 1% Propane and various other gases. Thus, statement 1 is wrong. Abundant coalbed methane sources exist, and also a huge deposit of shale gas is also found in the Indo-Gangetic plain, Assam, Gujarat, Rajasthan and the coastal region of India.

47. Consider the following problems being faced by the Indian coal industry-

I. Poor quality of coal and bottlenecks in the coal movement.

II. Low utilization capacity of washeries.

III. Growing dependence on the import of coking coal.

IV. Administered prices.

Which of the above are correct?

- (a) II, III and IV (b) I, II, III and IV
(c) I, III and IV (d) I, II and III

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

Currently, the coal industry in India is facing a lot of challenges which are low-quality coal, deficiencies in coal washing establishments, high imports of coking coal and administered prices, etc.

48. Maximum coal is used in –

- (a) Energy production (b) Iron and steel industry
(c) Railway (d) Steamships

U.P.P.C.S. (Mains) 2008

Ans. (a)

Coal plays a vital role in meeting energy needs worldwide. Coal is an indispensable energy source in India.

49. Which of the following do not have a major reserve of lignite?

- (a) Gujarat (b) West Bengal
(c) Jharkhand (d) Puducherry

U.P. R.O./A.R.O. (Mains) 2017

Ans. (c)

When the question was asked, option (c) was the correct answer. Lignite is popularly known as brown coal. It is formed due to the decay of vegetations metamorphism and other geological processes over thousands of years. Lignite is characterized by high moisture content, low ash content and low heating value.

According to the Indian Minerals Yearbook 2022, lignite reserves/resources (April 1, 2022) are found mainly in three states of : Tamil Nadu, Rajasthan and Gujarat.

50. Which one of the following river valleys is most important in respect of coal reserves of the Gondwana period?

- (a) The Son river valley (b) Mahanadi river valley
(c) Damodar river valley (d) Godavari river valley
(e) None of the above

Chhattisgarh P.C.S. (Pre) 2017

Ans. (c)

Damodar river valley is the largest in terms of reserves of Gondwana age coal. Jharia, Dhanbad and Bokaro of Jharkhand and the Raniganj coalfield of West Bengal come in the Damodar river valley. The Raniganj coalfield is also extended to Jharkhand.

51. Match List-I with List-II and select the correct answer from the codes given below :

List-I (Coalfields)	List-II (States)
A. Rajmahal	1. West Bengal
B. Sohagpur	2. Chhattisgarh
C. Bisrampur	3. Madhya Pradesh
D. Raniganj	4. Jharkhand

Code :

A	B	C	D
(a) 1	2	3	4
(b) 4	3	2	1
(c) 1	4	3	2
(d) 3	2	4	1

U.P. P.C.S. (Mains) 2017

Ans. (b)

(Coalfields)	(States)
Rajmahal	Jharkhand and Bihar
Sohagpur	Madhya Pradesh and Chhattisgarh
Bisrampur	Chhattisgarh
Raniganj	West Bengal and Jharkhand

52. Match List-I with List-II and select the correct answer from the code given below the lists:

List-I (Coal field)	List-II (Location)
A. Makum	1. Assam
B. Namchik	2. Arunachal Pradesh
C. Giridih	3. Jharkhand
D. Sohagpur	4. Madhya Pradesh

Code :

A	B	C	D
(a) 1	2	3	4
(b) 1	2	4	3
(c) 1	4	3	2
(d) 2	1	3	4

U.P. B.E.O. (Pre) 2019

Ans. (a)

Namchik - Namphuk coalfield is the only coalfield located in Changlang district in Arunachal Pradesh. Makum is a coalfield is located in Assam, Giridih in Jharkhand produces the best quality coal field. Other areas in Jharkhand, Bokaro, Jharia and Karanpura are also famous for coal production.

53. In which one of the following states are Namchik-Namphuk coal fields located ?

- Arunachal Pradesh
- Meghalaya
- Mizoram
- Manipur

I.A.S. (Pre) 2008

U.P.P.C.S. (Spl.) (Pre) 2008

Ans. (a)

The Namchik- Namphuk coal fields are situated in the Changlang district of Arunachal Pradesh.

54. Match List-I with List-II and select the correct answer from the code given below:

List-I (Coal field)	List-II (State of India)
A. Jhilmili	1. Odisha
B. Karanpura	2. Madhya Pradesh
C. Rampur Himgir	3. Chhattisgarh
D. Sohagpur	4. Jharkhand

Code :

A	B	C	D
(a) 3	4	2	1
(b) 1	2	4	3

- 2 1 3 4
- 3 4 1 2

U.P. R.O./A.R.O. (Mains) 2016

Ans. (d)

(Coal Field)	(Producer state)
Jhilmili	- Chhattisgarh
Karanpura	- Jharkhand
Rampur Himgir	- Odisha
Sohagpur	- Madhya Pradesh and Chhattisgarh

55. Match List-I with List-II and select the correct answer in the code given below:

List-I (Coal fields)	List-II (Indian States)
A. Ranjit Valley	1. Jharkhand
B. Tandur	2. Sikkim
C. Warora	3. Telangana
D. Lalmatia	4. Maharashtra

Code :

A	B	C	D
(a) 1	2	3	4
(b) 4	3	1	2
(c) 2	3	4	1
(d) 2	4	3	1

U.P. R.O./A.R.O. (Mains) 2016

Ans. (c)

Ranjit Valley	- Sikkim
Tandur	- Telangana
Warora	- Maharashtra
Lalmatia	- Jharkhand

56. Match List-I with List-II and choose the correct answer using the codes given below in the lists:

List-I (Coal Resources)	List-II (States)
A. Hutar	1. Madhya Pradesh
B. Ib River	2. Chhattisgarh
C. Tatapani - Ramkola	3. Odisha
D. Umaria	4. Jharkhand

Code :

A	B	C	D
(a) 3	2	4	1
(b) 2	1	3	4
(c) 4	3	2	1
(d) 4	2	3	1

UPPSC (Pre) 2024

Ans. (c)